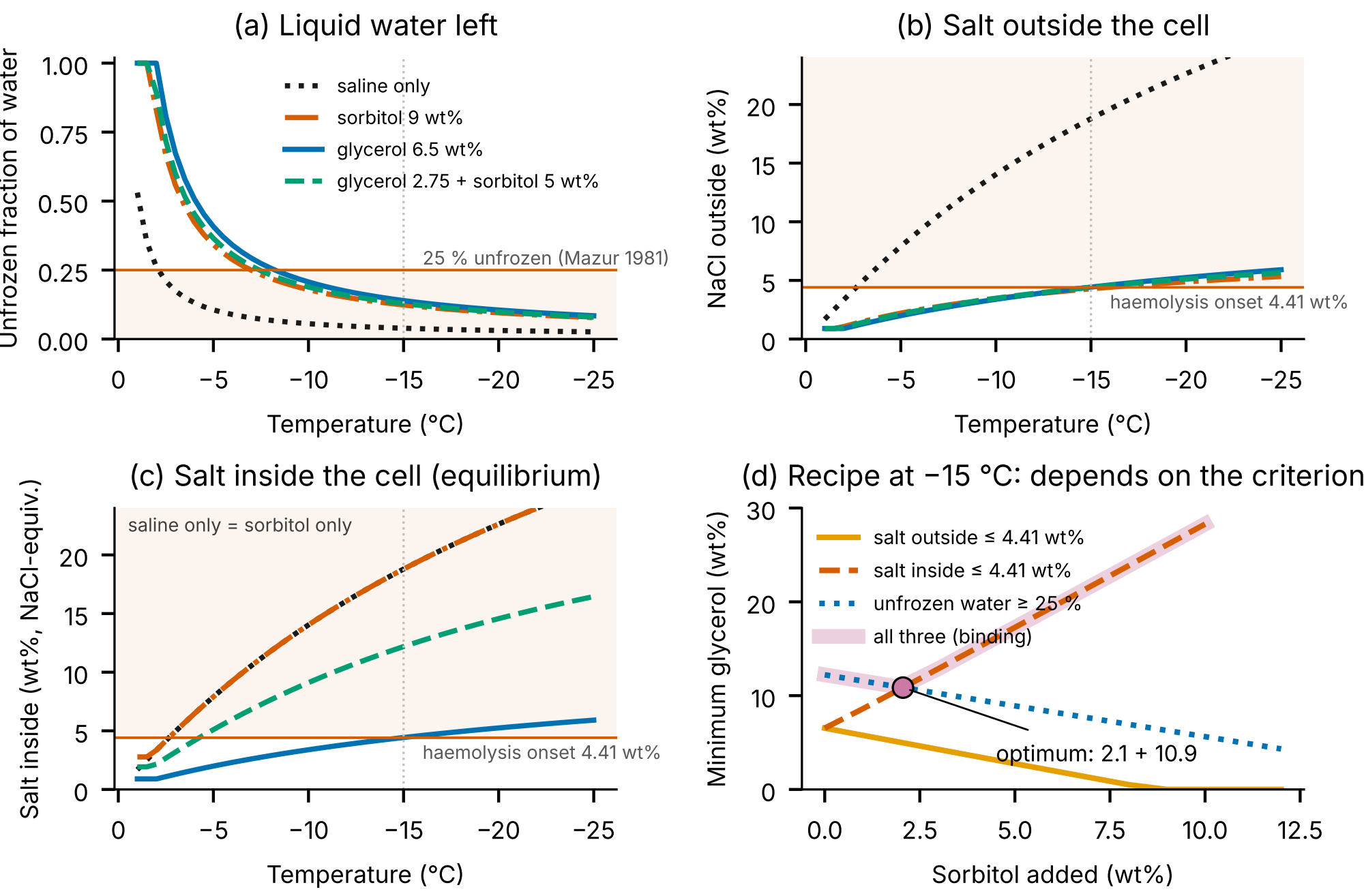
(a) Liquid water left
Unfrozen fraction of water
1.00
0.75
0.50
0.25
0.00
saline only
sorbitol 9 wt%
glycerol 6.5 wt%
glycerol 2.75 + sorbitol 5 wt%
25 % unfrozen (Mazur 1981)
0
−5
−10
−15
−20
−25
Temperature (°C)
(b) Salt outside the cell
NaCl outside (wt%)
20
15
10
5
0
haemolysis onset 4.41 wt%
0
−5
−10
−15
−20
−25
Temperature (°C)
(c) Salt inside the cell (equilibrium)
Salt inside (wt%, NaCl-equiv.)
saline only = sorbitol only
20
15
10
5
0
haemolysis onset 4.41 wt%
0
−5
−10
−15
−20
−25
Temperature (°C)
(d) Recipe at −15 °C: depends on the criterion
Minimum glycerol (wt%)
30
20
10
0
salt outside ≤ 4.41 wt%
salt inside ≤ 4.41 wt%
unfrozen water ≥ 25 %
all three (binding)
optimum: 2.1 + 10.9
0.0
2.5
5.0
7.5
10.0
12.5
Sorbitol added (wt%)